1. The Use Case (What We Are Solving)

Title: Autonomous Pre-Production Triage & Self-Learning RCA for CI/CD Pipelines

The Problem: In standard DevOps pipelines, when a build fails due to a security vulnerability or compliance violation, it creates a manual bottleneck. SREs and DevOps engineers suffer from massive "alert fatigue" because they have to manually read Jenkins logs, identify the error, write a Root Cause Analysis (RCA), create a Jira ticket, and notify the developer. Worse, they often repeat this manual process for the exact same errors, leading to duplicate tickets and wasted hours.

The Solution: An Agentic AI Middle Layer that autonomously intercepts Jenkins failures, checks a historical knowledge base (RAG) to see if the error is known, instantly halts the pipeline, and provides the exact fix to the developer via Slack—only escalating to a human for brand-new, unseen errors.

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2. Business Value (What It Will Increase)

- **Increases Developer Velocity:** Developers no longer wait hours for an SRE to triage a failed build. They receive an instant Slack message with the specific code fix within seconds of the pipeline failing.
- **Increases SRE / DevOps Capacity:** By handling 80% of routine pipeline failures (the "Known Issues"), highly paid engineers are freed up to focus on strategic architecture rather than repetitive debugging.
- **Increases Institutional Knowledge:** The Human-in-the-Loop (HITL) feedback cycle ensures that every time a new error is solved, the AI learns it permanently. The RAG database becomes an ever-growing, automated runbook.

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3. Process & Business Impact

Process Impact (How the workflow changes):

- **Zero Ticket Bloat:** The AI checks historical data *before* creating a ticket. Known errors update existing tickets instead of creating hundreds of duplicates, keeping the Jira backlog pristine.
- **True "Shift-Left" Resolution:** The AI halts the pipeline in the pre-production phase and forces the fix immediately, guaranteeing that insecure or non-compliant code never reaches the production environment.
- **Governed Automation:** The AI does the heavy lifting (drafting the RCA), but the process maintains strict governance. The system cannot add new rules to its memory without a human clicking "Approve" in Slack.

Business Impact (The bottom line):

- **Drastic MTTR Reduction:** Mean Time to Recovery drops from hours (manual log digging) to seconds (instant LLM log parsing).
- **Cost Avoidance:** Prevents costly security breaches and compliance fines by autonomously catching hardcoded secrets or policy violations before they are deployed.
- **Seamless Scalability:** As the engineering team grows and pushes more code, the AI Middle Layer scales infinitely to handle the increased pipeline volume without requiring the business to hire additional DevOps support staff.